

EVEN INTEGERS AS SUMS OF TWO SUMS OF TWO SQUARES WITH FEW PRIME FACTORS

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ABSTRACT. Let $N \equiv 2 \pmod{4}$ be sufficiently large. We show that the number of $n \leq N$ such that n and $N - n$ are both sums of two squares and $\Omega(n(N - n)) \leq 11$ is at least $\frac{3}{2} \mathfrak{S}(N)N/(\log N)^2$, where $\mathfrak{S}(N)$ is the binary Goldbach singular series. The level of distribution of the sequence $n(N - n)$ is 1, but sifting the primes $q \equiv 3 \pmod{4}$ up to $N^{1/2}$ with the linear sieve falls exactly on the sieving limit $s = 2$. We therefore sift these primes only up to $N^{1/2-\delta}$ and remove the survivors with two large prime factors $\equiv 3 \pmod{4}$ on the same side by a switching argument at the Bombieri–Vinogradov level. The final numerical inequality is verified with interval arithmetic.

1. INTRODUCTION

Every positive integer is a sum of four squares, so every N is of the form $n_1 + n_2$ with n_1, n_2 sums of two squares. We ask for such representations in which n_1 and n_2 have few prime factors.

The multiplicative analogue, pairs $n, n + 1$ that are both sums of two squares, was settled up to constants by Hooley [5] and Indlekofer [6], who showed that there are $\asymp x/\log x$ such $n \leq x$; correlations $\sum r(n)r(n + h)$ of the representation function have since been studied by spectral methods, with uniformity in the coefficients of the forms, e.g. by Blomer [1]. On the side of almost prime variables, Brüdern and Fouvry [3] proved Lagrange’s four squares theorem with almost prime variables, and Blomer, Grimmelt, Li and Rydin Myerson [2] showed that every large $n \equiv 1, 3 \pmod{6}$ is a prime plus two squares of almost primes. In these works the almost prime condition is on the variables x_i . Here it is on the values $x^2 + y^2$ themselves. Finally, [4] treats the case where one summand is a prime and the other is an almost prime sum of two squares; the present problem is its two-sided analogue.

Theorem 1.1. *For every sufficiently large $N \equiv 2 \pmod{4}$,*

$$\begin{aligned} \#\{n \leq N : n \equiv 1 \pmod{4}, n = x^2 + y^2, N - n = z^2 + w^2, \Omega(n(N - n)) \leq 11\} \\ \geq \frac{3}{2} \mathfrak{S}(N) \frac{N}{(\log N)^2}, \end{aligned}$$

where $\mathfrak{S}(N) = 2 \prod_{\ell > 2} (1 - (\ell - 1)^{-2}) \prod_{\ell | N, \ell > 2} \frac{\ell - 1}{\ell - 2}$.

Remark 1.2 (The prime 2). The restriction $N \equiv 2 \pmod{4}$ is natural. The odd part of a sum of two squares is $\equiv 1 \pmod{4}$. If $N \equiv 0 \pmod{4}$, both summands must contain powers of 2, and for $N = 2^j$ the only representation is $n_1 = n_2 = 2^{j-1}$, so $\Omega(n_1 n_2)$ is unbounded. For general N one should count prime factors of the odd parts only; we do not pursue this here.

Remark 1.3. With the parameters of the second line of Table 1 the same method gives the weaker bound $\Omega(n(N - n)) \leq 12$ with the larger constant 2.24 in place of $\frac{3}{2}$; both are certified.

Method. Restrict to $n \equiv 1 \pmod{4}$; then $N - n \equiv 1 \pmod{4}$. If neither n nor $N - n$ has a prime factor $q \equiv 3 \pmod{4}$, both are sums of two squares. The sequence $\{n(N - n)\}$ has level of distribution $N^{1-\varepsilon}$ (Lemma 2.1), and the primes $q \equiv 3 \pmod{4}$ have two roots of $n(N - n) \equiv 0$

(mod q), so they form a sifting set of dimension 1. Sifting them up to $N^{1/2}$ at level N gives $s = 2$, where the linear lower bound function vanishes. We sift them up to $y = N^{1/2-\delta}$ instead. Since $n \equiv N - n \equiv 1 \pmod{4}$, each side has an even number of such prime factors, so a survivor that is not a pair of sums of two squares has two large prime factors $\equiv 3 \pmod{4}$ on the *same* side; these are counted from above by switching (Section 3). Prime factors are controlled by Richert's weights applied to $n(N - n)$ jointly.

Remark 1.4. To the best of the author's knowledge, Theorem 1.1 has not appeared in the literature.

2. THE SEQUENCE $n(N - n)$

Let $N \equiv 2 \pmod{4}$ be large, $\mathcal{P}_1 = \{q \equiv 1 \pmod{4}\}$, $\mathcal{P}_3 = \{q \equiv 3 \pmod{4}\}$, and

$$\mathcal{A} = \{a_n = n(N - n) : 1 \leq n \leq N, n \equiv 1 \pmod{4}\}, \quad X = N/4.$$

For odd squarefree d let $\rho(d)$ be the number of solutions of $n(N - n) \equiv 0 \pmod{d}$; it is multiplicative with $\rho(q) = 2$ for $q \nmid N$ and $\rho(q) = 1$ for $q \mid N$. Put $g(d) = \rho(d)/d$, so that

$$g(q) = \frac{2}{q} \quad (q \nmid N), \quad g(q) = \frac{1}{q} \quad (q \mid N),$$

and define r_d by $\#\{n : a_n \in \mathcal{A}, d \mid a_n\} = Xg(d) + r_d$.

Lemma 2.1 (Level of distribution). $|r_d| \leq \rho(d)$, and for $D \leq N$, $\sum_{d \leq D} \mu^2(d) \tau_3(d) |r_d| \ll D(\log D)^C$. In particular the level of distribution is $N^{1-\varepsilon}$.

Proof. The condition $d \mid a_n$ together with $n \equiv 1 \pmod{4}$ restricts n to $\rho(d)$ classes modulo $4d$, each containing $N/(4d) + O(1)$ integers $n \leq N$. Hence $|r_d| \leq \rho(d) \leq 2^{\omega(d)}$, and the sum is $\ll \sum_{d \leq D} \tau_3(d) 2^{\omega(d)} \ll D(\log D)^C$. $\square$

Lemma 2.2 (Local factors). As $z \rightarrow \infty$,

$$\prod_{2 < q < z} (1 - g(q)) \sim \frac{\mathfrak{S}(N)}{2} \left(\frac{2e^{-\gamma}}{\log z} \right)^2.$$

Proof. For $q \nmid N$, $(1 - 2/q)(1 - 1/q)^{-2} = 1 - (q - 1)^{-2}$; for $q \mid N$, $(1 - 1/q)(1 - 1/q)^{-2} = q/(q - 1)$. The product of these factors over odd q converges to $\mathfrak{S}(N)/2$, and $\prod_{2 < q < z} (1 - 1/q) \sim 2e^{-\gamma}/\log z$. $\square$

We use the following sieve tools; see [4] for statements and references. The linear sieve ($\kappa = 1$, $\beta = 2$) has functions F_1, f_1 with $F_1(s) = 2e^\gamma/s$ for $0 < s \leq 3$ and $f_1(s) = 2e^\gamma \log(s - 1)/s$ for $2 \leq s \leq 4$, $f_1(s) = 0$ for $s \leq 2$ [8]; the semi-linear sieve ($\kappa = 1/2$, $\beta = 1$) has functions $F_{1/2}, f_{1/2}$ [7]. The vector sieve inequality [3] is $ab \geq a^-b^+ + a^+b^- - a^+b^+$ for $a, b \in \{0, 1\}$, $a^- \leq a \leq a^+$, $b^- \leq b \leq b^+$, $a^+, b^+ \geq 0$. The Bombieri–Vinogradov theorem for convolutions is used as in [4, Lemma 2.4].

3. THE WEIGHTED SIEVE AND SWITCHING

Fix $0 < \alpha < \delta < 1/4$, $0 < \lambda$, $\alpha < \beta_1 \leq 1/2$, and put

$$y = N^{1/2-\delta}, \quad z = N^\alpha, \quad y_1 = N^{\beta_1}, \quad P_I = \prod_{\substack{q \in \mathcal{P}_3 \\ q < y}} q, \quad P_S = \prod_{\substack{q \in \mathcal{P}_1 \\ q < z}} q, \quad P = P_I P_S.$$

Every odd prime $q < z$ divides P . For $a_n \in \mathcal{A}$ with $(a_n, P) = 1$ define

$$w_n = 1 - \lambda \sum_{\substack{q|a_n, q \in \mathcal{P}_1 \\ z \leq q < y_1}} \left(1 - \frac{\log q}{\log y_1}\right), \quad W = \sum_{(a_n, P)=1} w_n.$$

Lemma 3.1 (Structure). *Let $(a_n, P) = 1$. Then either*

- (i) *n and $N - n$ are both sums of two squares, all their prime factors are $\geq z$, and those in $\mathcal{P}_3$ occur only as squares q^2 ; or*
- (ii) *for $m' = n$ or $m' = N - n$ we have $m' = q_1 q_2 m$ with $q_1 < q_2$, $q_1, q_2 \in \mathcal{P}_3$, $q_1 \geq y$, and $m \leq N^{2\delta}$ composed of primes of $\mathcal{P}_1$ that are $\geq z$.*

Proof. Each of n , $N - n$ is odd and $\equiv 1 \pmod{4}$, so each has an even number of prime factors in $\mathcal{P}_3$, counted with multiplicity. They are all $\geq y$, and $y^4 > N$, so each side has 0 or 2 of them. If on both sides there are none, or they coincide, both sides are sums of two squares. Otherwise some side is as in (ii). In particular, two exceptional prime factors never lie on different sides. $\square$

Lemma 3.2 (Prime factors). *If a_n is as in Lemma 3.1(i), squarefree, and $w_n > 0$, then $\Omega(n(N - n)) < 1/\lambda + 2/\beta_1$.*

Proof. All prime factors of a_n are in $\mathcal{P}_1$ and $\geq z$, and $\log a_n \leq 2 \log N$. As in [4, Lemma 3.2], $\sum_{q|a_n, q < y_1} (1 - \log q / \log y_1) \geq \Omega(a_n) - \log a_n / \log y_1 \geq \Omega(a_n) - 2/\beta_1$. $\square$

Non-squarefree a_n with a square factor q^2 , $q \geq z$, number $\ll N^{1-\alpha}$ and are negligible.

Proposition 3.3 (Main term). *Let $\theta_I \in (0, 1)$ and, for $u \in [\alpha, \beta_1]$, $t(u) \in (0, 1 - u)$. Put $s_I = \theta_I / (1/2 - \delta)$, $s_S = (1 - \theta_I) / \alpha$,*

$$L = f_1(s_I)F_1(s_S) + F_1(s_I)f_1(s_S) - F_1(s_I)F_1(s_S),$$

$$R = \int_{\alpha}^{\beta_1} \left(1 - \frac{u}{\beta_1}\right) F_1\left(\frac{t(u)}{1/2 - \delta}\right) F_1\left(\frac{1 - u - t(u)}{\alpha}\right) \frac{du}{u}.$$

Then $W \geq (L - \lambda R + o(1)) XV$, where $V = \prod_{q|P} (1 - g(q))$ and

$$XV \sim \frac{e^{-2\gamma}}{2\alpha(1/2 - \delta)} \mathfrak{S}(N) \frac{N}{(\log N)^2}.$$

Proof. Apply the vector sieve with $a = 1_{(a_n, P_I)=1}$ and $b = 1_{(a_n, P_S)=1}$. Both are linear sieves: P_I and P_S consist of primes from one half of the residue classes modulo 4, each with $\rho(q) = 2$ (apart from the finitely many $q \mid N$ below a given bound, which do not affect the dimension). Use levels N^{θ_I} and $N^{1-\theta_I-\varepsilon}$; the combined moduli are $\leq N^{1-\varepsilon}$ and Lemma 2.1 controls the remainders. For the weights, $q \mid a_n$ for $q \in \mathcal{P}_1$, $q \nmid N$, has density $g(q) = 2/q$, and $\sum_{q \in \mathcal{P}_1} 2/q$ over $N^u \leq q < N^{u+du}$ is du/u . The upper bound vector sieve for $\mathcal{A}_q$ at levels $N^{t(u)}$, $N^{1-u-t(u)-\varepsilon}$ gives R . Finally $V = \prod_{2 < q < z} (1 - g(q)) \prod_{q \in \mathcal{P}_3, z \leq q < y} (1 - g(q))$, and Lemma 2.2 together with $\prod_{q \in \mathcal{P}_3, z \leq q < y} (1 - 2/q) \sim \log z / \log y$ gives the asymptotic for XV . $\square$

Proposition 3.4 (Switching). *Let $\mathcal{E}$ be the set of n with $(a_n, P) = 1$ in case (ii) of Lemma 3.1. Then*

$$\#\mathcal{E} \leq \left(4c(\delta, \alpha) e^{\gamma} \sqrt{\alpha(1/2 - \delta)} U + o(1)\right) XV,$$

where $c(\delta, \alpha)$ is the switching density of [4, Prop. 4.1] and

$$U = \min_{0 < t < 1/2} F_{1/2}\left(\frac{t}{1/2 - \delta}\right) F_{1/2}\left(\frac{1/2 - t}{\alpha}\right).$$

Proof. Write $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2$ according as the exceptional pair lies in n or in $N - n$; the two cases are symmetric, which accounts for a factor 2. Consider $\mathcal{E}_1$. Let $\mathcal{T}$ be the set of triples (q_1, q_2, m) as in Lemma 3.1(ii) with $t = q_1 q_2 m \leq N$, so that $|\mathcal{T}| = (c(\delta, \alpha) + o(1))N/\log N$ as in [4]; note $t \equiv 1 \pmod{4}$ automatically. Every $n \in \mathcal{E}_1$ equals such a t , and $N - t$ has no prime factor in P . Hence $\#\mathcal{E}_1$ is at most the number of elements of $\mathcal{B} = \{N - t : t \in \mathcal{T}\}$ coprime to P .

Let $\ell \mid P$. No prime factor of t divides P (the q_i are $\geq y$ and the prime factors of m are split and $\geq z$), so t is equidistributed in the nonzero classes modulo ℓ . Hence, for $\ell \nmid N$, $\ell \mid N - t$ with density $g_{\mathcal{B}}(\ell) = 1/(\ell - 1)$, while for $\ell \mid N$ we have $\ell \nmid N - t$, so $g_{\mathcal{B}}(\ell) = 0$. The upper bound vector sieve for $\mathcal{B}$ with the semi-linear components P_I and P_S at total level $N^{1/2}(\log N)^{-B}$ (Bombieri–Vinogradov for the convolution $q_1 * (q_2 m)$) gives

$$\#\mathcal{E}_1 \leq (U + o(1)) |\mathcal{B}| \prod_{q \mid P} (1 - g_{\mathcal{B}}(q)).$$

Now $(1 - 1/(\ell - 1))(1 - 1/\ell)^{-1} = 1 - (\ell - 1)^{-2}$ and $(1 - 0)(1 - 1/\ell)^{-1} = \ell/(\ell - 1)$ for $\ell \mid N$: these are the same local factors as in Lemma 2.2. Hence

$$\prod_{q \mid P} (1 - g_{\mathcal{B}}(q)) \sim \frac{\mathfrak{S}(N)}{2} \cdot \frac{2e^{-\gamma}}{\log z} \left(\frac{\log z}{\log y} \right)^{1/2},$$

and therefore

$$\#\mathcal{E}_1 \leq (U + o(1)) c \frac{N}{\log N} \cdot \frac{\mathfrak{S}(N)e^{-\gamma}}{\alpha \log N} \left(\frac{\alpha}{1/2 - \delta} \right)^{1/2} = (2ce^{\gamma} \sqrt{\alpha(1/2 - \delta)} U + o(1)) XV,$$

using Proposition 3.3 for XV . The singular series cancels exactly. Adding $\mathcal{E}_2$ gives the claim. $\square$

4. NUMERICAL VERIFICATION AND PROOF OF THEOREM 1.1

By Lemma 3.1 and $w_n \leq 1$, the weighted count over case (i) is at least $W - \#\mathcal{E} \geq (\text{net} + o(1))XV$ with

$$\text{net} = L - \lambda R - 4c(\delta, \alpha)e^{\gamma} \sqrt{\alpha(1/2 - \delta)} U.$$

By Lemma 3.2, every squarefree counted n with $w_n > 0$ satisfies $\Omega(n(N - n)) < 1/\lambda + 2/\beta_1$. We use the two parameter sets of Table 1; in both, $1/\lambda + 2/\beta_1$ is an integer, computed in exact rational arithmetic.

δ	α	λ	β_1	k	$L \geq$	$R \leq$	$c \leq$	$U \leq$	net $\geq$
0.245	0.016	1/8	1/2	11	0.944016	2.830154	1.067736	1.134620	0.038949
0.22	0.010	1/8	2/5	12	0.913373	3.144708	1.092388	1.166735	0.039811

TABLE 1. Certified bounds. Here $k = \lceil 1/\lambda + 2/\beta_1 \rceil - 1$; the linear sieve level split is $\theta_I = 0.9220$ resp. 0.9495.

Lemma 4.1. *The bounds of Table 1 hold.*

Method of verification. The computation is carried out by the script `code/certify.py`.

- (1) The sieve functions F_1, f_1 ($\kappa = 1$) and $F_{1/2}, f_{1/2}$ ($\kappa = 1/2$) are evaluated from their closed forms on $(0, 3]$, $[2, 4]$ resp. $(0, 2]$, $[1, 3]$ in interval arithmetic (`mpmath.iv`, 80 bits). Beyond these ranges the delay-differential equations $(s^{\kappa} F(s))' = \kappa s^{\kappa-1} f(s - 1)$, $(s^{\kappa} f(s))' = \kappa s^{\kappa-1} F(s - 1)$ are integrated on the grid $10^{-3}\mathbb{Z}$ (up to $s = 110$ resp. 60): on each step the lagged function is enclosed between its values at the end points, using that F decreases and f increases, and $\int \kappa s^{\kappa-1} ds$ is computed exactly. Each floating point

operation is followed by one outward rounding step. Off the grid, monotonicity gives one-sided bounds.

- (2) L is bounded below via $L = f_1(s_I)F_1(s_S) + F_1(s_I)(f_1(s_S) - F_1(s_S))$ and $f_1(s_S) - F_1(s_S) \leq 0$.
- (3) R is bounded above by a Riemann sum over 4000 cells with $t(u)$ constant on each cell; on a cell $[u_a, u_b]$ the factor $(1 - u/\beta_1)/u$ is largest at u_a and $F_1((1 - u - t)/\alpha)$ is largest at u_b . U is bounded above by evaluating the product at one admissible t .
- (4) For c , the measure $\frac{1}{2} 1_{[\alpha, 2\delta]}(\mu) d\mu/\mu$ is replaced by the masses $\frac{1}{2} \log \frac{k+1}{k}$ placed at the left end points $k/2000$. The integrand $I(\mu) = \frac{1}{1-\mu} \log \frac{1/2+\delta-\mu}{1/2-\delta}$ is decreasing on $[0, 2\delta]$: its derivative has the sign of $\log \frac{a-\mu}{b} - \frac{1-\mu}{a-\mu}$ with $a = \frac{1}{2} + \delta < 1$, $b = \frac{1}{2} - \delta$, whose first term decreases and second increases, and which is negative at $\mu = 0$ (checked in the script). So moving mass to the left gives an upper bound. The floating point convolution is multiplied by $1 + 10^{-9}$.

A finer independent computation of $c(0.22, 0.01)$ (grid $1/20000$) gives 1.0913, consistent with the certified upper bound. $\square$

Proof of Theorem 1.1. Take the first line of Table 1. By Lemma 4.1 and Proposition 3.3, after removing the $\ll N^{1-\alpha}$ non-squarefree a_n , the number of counted n is at least

$$(0.038949 + o(1)) \frac{e^{-2\gamma}}{2 \cdot 0.016 \cdot 0.255} \mathfrak{S}(N) \frac{N}{(\log N)^2} \geq \frac{3}{2} \mathfrak{S}(N) \frac{N}{(\log N)^2}$$

for large N (the numerical constant is ≥ 1.5046). Each of them has n and $N - n$ sums of two squares and $\Omega(n(N - n)) \leq 11$. $\square$

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